Structural Calibration of the Persuasive-Recommendation Model on the MIND Dataset

Hossein Piri

August 2026

This note takes Jiangze's recommendation-persuasion model, and the finite-horizon learning version in Meichun's note, to the MIND data. It estimates the model's two primitives, the consumption function and the taste transition, and then simulates cultivation-aware and cultivation-blind policies in the calibrated environment. Three findings organize the note. Alignment between a user's taste state and an item's category is a first-order driver of clicks. Recent consumption updates an effective preference state that significantly improves next-day click prediction, though the data do not separate causal taste change from persistent interests, exposure adaptation, or the platform learning a stable profile. And forward-looking planning adds almost nothing beyond reactive myopia in any scenario anchored to the calibrated parameters: uniform or adversarially chosen margins, one-hot or dense content geometry, short or long horizons. The one combination that produces a real planning premium pairs the resistance term the data themselves motivate with alignment sensitivity above a quantified gate, which turns the null into a sharp statement about where cultivation economics live.

1 Mapping the model to MIND

The model has three primitives: a taste state z, a consumption probability $p_i(z) = \sigma(\alpha_i + \kappa\, z^\top x_i)$, and a transition $z^+ = (1-\rho)z + \rho x_i$ that fires only on consumption. MIND supplies each object at the category level. The taste state is a user's distribution over the 18 top-level news categories; item features are category indicators, $x_i = e_{c(i)}$, so the alignment $z^\top x_i$ is the user's history share of the item's category; an impression is a take-it-or-leave-it recommendation and a click is consumption. The empirical actions are therefore persistent content categories rather than individual articles, which matters for the learning interpretation below.

Two identification choices matter. First, the state entering the consumption model is built only from the *history* field, clicks that predate the log window, so alignment is predetermined at the time of the impression. Second, the transition parameter is identified temporally: z evolves through clicks during the training window (November 9–14) and is scored on next-day click behavior (November 15). A running-average taste built from the same clicks it predicts would correlate with ρ mechanically; separating the window that moves the state from the window that evaluates it removes that circularity.

MIND has no revenue data. The simulation's baseline sets $m_i = 1$ and measures value in expected clicks; the margin scenarios in Section 4 use illustrative advertising-value ratios and are counterfactual calibrations, not estimates.

2 The consumption function

The first question is whether the logistic-in-alignment consumption function describes click behavior. We fit $P(\text{click}) = \sigma(\alpha_{c(i)} + \kappa\, z^\top x_i)$ by maximum likelihood on 1,999,923 impression–item observations (a random sample of the training window; users with at least five history clicks; click rate 4.1%), with 999,975 held-out observations for validation.

The alignment coefficient is $\hat{\kappa} = 1.751$ (SE 0.021; user-clustered SE ≈ 0.026), with a likelihood ratio of 6,539 against the no-alignment null. Alignment is not a marginal refinement of category popularity; it is a first-order driver of consumption. The robustness checks all push in the conservative direction: 211 subcategory effects in place of the 14 category effects move $\hat{\kappa}$ to 1.78, user-activity controls raise it, and a within-user fixed-effects specification raises the implied coefficient further, so population heterogeneity attenuates rather than manufactures the effect. One qualification bounds what $\hat{\kappa}$ means: it is a behavioral click coefficient under MSN's incumbent exposure policy, not a causal taste primitive.

Because alignment scales differ across feature representations, the comparable object is the standardized effect, κ times the standard deviation of alignment, and it is stable. The binary logit gives 0.33 log-odds per SD with category features and 0.34 with dense LSA article embeddings (Section 4 introduces these); adding position controls leaves it at 0.34. Within-impression competition, the concern that a nonclick means another item was chosen rather than a decline, moves the estimate the other way: a slate-level conditional logit with position-bucket effects, fit on the 71% of impressions with exactly one click (every MIND train impression contains a click, so the model conditions on a click occurring), *raises* the category coefficient to 2.75 (0.51 per SD; 0.36 dense) by identifying alignment from within-impression contrasts. In the binary model position effects are strong and monotone, reaching -1.43 log-odds beyond position twenty; the conditional logit differences most of this out, identifying it as slate-length dilution rather than pure attention decay. Every specification error found so far biases $\hat{\kappa}$ toward zero.

Holdout AUC is 0.612, against 0.707 for our full-feature MNL. The gap of ten AUC points is the predictive price of compressing every article to its category, and it belongs in the paper: the structural model buys interpretability with real predictive currency.

Table 1 reports the category intercepts, the parameters the platform learns in Meichun's learning analysis. Four of the 18 categories (games, kids, middleeast, northamerica) have too few impressions to carry an intercept and are dropped from estimation. The intercepts are tightly estimated and heterogeneous, spanning 0.85 log-odds units, and no category's intercept advantage is large relative to alignment: moving a user's alignment from 0 to 0.5 changes log-odds by $\hat{\kappa}/2 \approx 0.88$, comparable to the entire intercept range. Cultivating alignment and picking high-α categories are forces of the same magnitude; Section 4 asks whether exploiting the former dynamically is worth anything.

3 The taste transition

The transition is the model's distinctive ingredient, and the simulation needs to know how fast the state actually moves. We estimate ρ by profile likelihood: for each ρ on a grid, every user's state is rolled forward through their training-window clicks, $z \leftarrow (1-\rho)z + \rho e_c$, and the implied alignment enters a logit of the 7,132,710 dev-window impressions of 120,000 users, refitting (α, κ) at each grid point. The nested null $\rho = 0$ says the state never moves.

Figure 1 shows the result: $\hat{\rho} = 0.035$, likelihood ratio 1,270.0 against $\rho = 0$. Three robustness layers, run at scale on the ARC cluster, pin this estimate down. First, invariance: re-estimating on the exact eight-category-plus-other state the simulator uses, and fixing κ at either the training-

Table 1: Category intercepts $\hat{\alpha}_c$ (logit scale).

Category	$\hat{\alpha}_c$	SE	n (fit)
music	-2.869	0.014	91,328
tv	-2.973	0.015	83,036
weather	-2.982	0.026	30,342
video	-3.124	0.027	32,107
lifestyle	-3.380	0.011	225,229
sports	-3.407	0.012	202,261
health	-3.442	0.017	103,351
finance	-3.448	0.012	192,542
movies	-3.494	0.027	44,712
entertainment	-3.567	0.017	119,985
foodanddrink	-3.684	0.017	127,571
news	-3.702	0.011	546,850
travel	-3.716	0.019	108,197
autos	-3.719	0.020	92,412

window value (1.751) or the dev-window value (2.141), returns $\hat{\rho} = 0.035$ in every one of six configurations, so the transition estimate does not depend on the category universe or on which κ it is paired with. Second, inference that respects user dependence: a user-block bootstrap (300 replicates, every state rebuilt inside each resample) gives a 95% interval of [0.030, 0.035], endpoints quantized at the 0.005 grid step, with every replicate rejecting $\rho = 0$. Third, a permutation placebo (dev click labels shuffled within user) collapses the likelihood ratio to noise, so the effect is identified by which items a user clicks, not by user-level composition. At $\hat{\rho}$ the dev-window alignment coefficient is 2.141 (SE 0.011), consistent with the training-window estimate. Two conversions make the magnitude concrete: a step of 0.035 implies a half-life of about 19 successful clicks, and after the average user's 7.6 in-window clicks the window's consumption carries roughly a quarter of the state weight.

What $\hat{\rho}$ means requires care, because four mechanisms generate the same signal over a 14-day window: true preference change, persistent short-run interest shocks, the incumbent recommender adapting next-day exposure to recent clicks, and the platform measuring a stable preference more precisely. The exposure channel is demonstrably active: a multinomial model of what MSN *shows* on the dev day, ignoring click labels entirely, also profiles in ρ, at least as strongly as clicking does. The honest statement is that recent consumption updates an effective predictive state under the incumbent policy; separating drift from measurement needs longer panels or exogenous exposure variation.

Two specification checks sharpen the transition, and both point to model extensions rather than confirmations. First, the model says the state moves only on consumption. Adding a separate update ρ_{ns} for shown-not-clicked items, evaluated at $\hat{\rho}$, yields $\hat{\rho}_{ns} = -0.015$ (likelihood ratio 520.3): exposure without a click is negatively associated with subsequent affinity for that category, a backfire effect. Three caveats keep this result at arm's length from the theory. Shown items are subsampled one-in-ten, so the per-actual-impression step is roughly ten times smaller and the structural magnitude is sampling-convention-dependent; a shown-but-unclicked article is not necessarily a declined offer, since the user may never have seen it; and on a coarse joint grid the click and nonclick updates partially confound each other, so the profile at $\hat{\rho}$ is the cleaner read. The

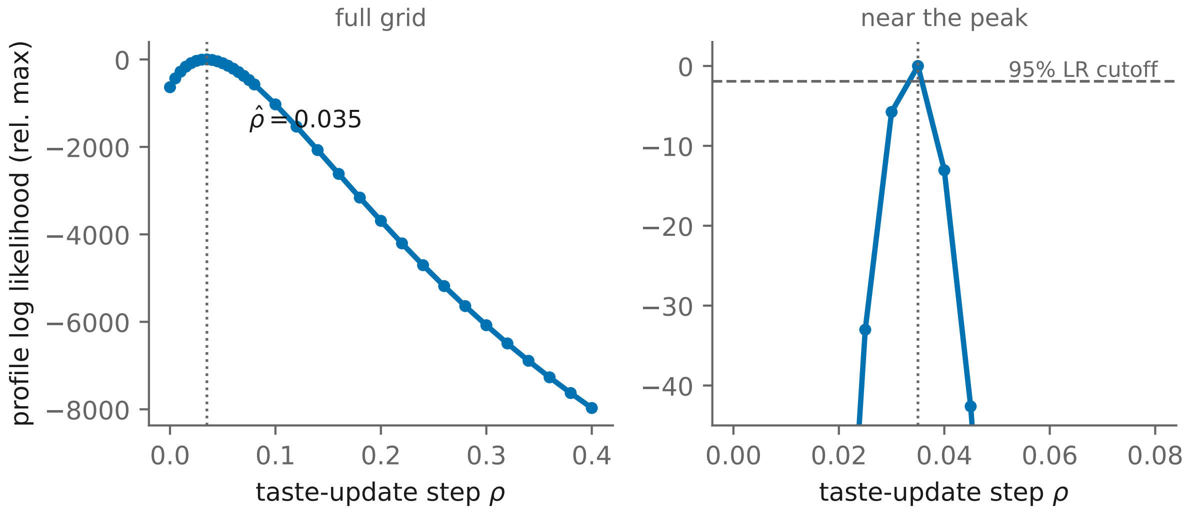

Figure 1: Profile log likelihood over ρ. The dashed line is the 95% likelihood-ratio cutoff.

sign, however, is unambiguous, and a negative nonclick update is exactly the resistance ingredient the model currently lacks.

Second, the constant-ρ update is tested against a diminishing-weight alternative in which each click carries weight $1/(n_0 + t)$. The running mean fits better, by 126.1 log-likelihood points at $n_0 = 20$, and the reason is systematic: the best-fit constant ρ falls with a user's history length almost exactly as $1/(N + T/2)$, the signature of Bayesian posterior updating of a stable taste. For calibration this is benign, since over this window the two updates prescribe nearly identical dynamics and $\hat{\rho} = 0.035$ is the right simulation input either way. For the model it is the more interesting fact: the effective update rate declines with tenure, so whatever cultivation is possible is concentrated among new and low-history users. Our five-click history filter excludes exactly the users for whom updating should be strongest, which makes the tenure gradient a lower bound and a hierarchical treatment of low-history users the natural follow-up.

Third, the update transfers only weakly to dense content space, and measuring this correctly requires care with norms. Repeating the profile with 50-dimensional LSA article embeddings, the raw update $(1-\rho)z + \rho x$ shrinks $\|z\|$ heterogeneously across users and mechanically penalizes every $\rho > 0$; our first pass mistook this artifact for a null, and the independent re-analysis caught it. With the state renormalized to unit length, stepping toward clicked articles' vectors peaks at $\rho = 0.02$ with a likelihood ratio of 68 on 30,000 users, decisively nonzero but roughly a fifth of the categorical evidence on comparable data, while stepping toward the clicked category's centroid remains flat at any ρ. The reading: most of the short-run predictive movement is categorical recency, a modest article-level residue exists on top of it, and the dense history profile otherwise behaves like a stable preference. Dense-geometry simulations should treat ρ as an imported sensitivity parameter.

4 The value of cultivation-aware planning

The claim the simulation tests is that a policy ignoring taste dynamics should perform materially worse than one planning through them. The calibrated environment uses the $n = 8$ largest categories with their estimated intercepts, $\kappa = 1.751$, horizon $H = 30$, initial states drawn from the empirical distribution of MIND user histories renormalized to the eight simulated categories, and ρ swept over a grid containing $\hat{\rho}$. Four policies cross knowledge with planning. Oracle policies know all parameters; learning policies know (κ, ρ) and learn the category intercepts across episodes by maximum likelihood, the estimate-then-plug-in benchmark of Meichun's learning analysis. Aware policies plan through the transition with depth-2 lookahead and a fluid rollout continuation; against exact dynamic programming this planner matched the optimal value on every discriminating instance in a 52-instance search, including instances where dynamic planning beats myopic by 28%, with a maximum relative value error of 0.7% on the hardest margin-misaligned validation, so it is a cultivation-aware rollout policy whose gaps read as slight underestimates of the true dynamic premium. Blind policies are reactive myopic: they observe the current state each period and take the best immediate action, ignoring how today's action moves tomorrow's state. All policies face common random numbers, so at $\rho = 0$ aware and blind coincide by construction.

4.1 The calibrated null

Two calibration refinements precede the results. Re-estimating jointly on the exact simulator state (eight categories plus an explicit other bucket, no renormalization) gives $\kappa = 1.966$, between the training and dev values, and every main scenario below was run at both $\kappa = 1.751$ and $\kappa = 2.141$ with per-user paired differences under common random numbers. The paired design supports an equivalence statement rather than a bare null: in every one of the 24 main cells, the cell's own one-sided 95% upper bound (t_9) on the aware-over-myopic premium is at most 0.53%; a Bonferroni-simultaneous version of the same statement sits near 0.6% and supports the same conclusion.

Table 2 reports the uniform-margin baseline, and the result is a null we did not script: *cultivation blindness costs essentially nothing at the calibrated parameters.* At $\hat{\rho} = 0.035$ the aware policy earns 2.01 clicks per user against 2.01 for the myopic policy, and the gap stays economically negligible over the entire grid, never exceeding 0.35% even at $\rho = 0.35$, ten times the estimated speed. The aware policy departs from the myopic action in only 0.2% of periods at $\hat{\rho}$. Expected clicks do rise with ρ for every policy: taste dynamics create value through self-reinforcement, clicks raising the clicked category's future click probability, but a reactive myopic policy harvests that value for free.

Table 2: Uniform margins: expected clicks per user (H=30, 3,000 users per cell, 5 seeds). The last column is the aware−blind gap among oracle policies; every gap is economically negligible.

ρ	oracle-aware	oracle-blind	learn-aware	learn-blind	gap
0.000	1.96	1.96	1.90	1.90	0.0%
0.035	2.01	2.01	1.96	1.96	0.0%
0.050	2.02	2.02	1.97	1.98	0.0%
0.150	2.17	2.17	2.10	2.11	0.0%
0.250	2.36	2.36	2.30	2.31	0.1%
0.350	2.50	2.49	2.41	2.43	0.3%

Margin heterogeneity does not overturn the null, and this is no longer a statement about one configuration. Table 3 shows the illustrative advertising-value margins (news 1.0 up to finance 4.0) producing at most 0.5% at $\rho = 0.35$; extending the horizon to $H = 120$ yields 0.9% at $\rho = 0.15$; a high-engagement counterfactual that raises every intercept by 3 log-odds units still produces gaps of at most 0.15%. A systematic sweep then searches the margin space itself: 93 margin vectors spanning three dispersion levels plus adversarial constructions with margins rank-reversed against category attractiveness and against popularity. No vector in the sweep exceeds 0.83% at $\rho = 0.15$, and that figure is itself an upward-biased order statistic over noisy estimates: fresh-seed replication of the best vector gives about 0.5%, and the best vectors fall to 0.17% at $\hat{\rho}$. The median gap is 0.03%, and the gap is uncorrelated with how adversarially margins are aligned against tastes. No margin configuration makes cultivation-aware planning worth having at the calibrated alignment sensitivity; the binding constraint sits elsewhere.

Table 3: Heterogeneous margins: expected revenue per user (H=30, 3,000 users per cell, 5 seeds).

ρ	oracle-aware	oracle-blind	learn-aware	learn-blind	gap
0.000	4.46	4.46	4.07	4.07	0.0%
0.035	4.55	4.55	4.11	4.10	0.0%
0.050	4.65	4.65	4.21	4.24	0.0%
0.150	5.10	5.09	4.55	4.65	0.1%
0.250	5.65	5.64	5.00	5.08	0.3%
0.350	6.19	6.16	5.50	5.58	0.5%

4.2 Why myopic keeps winning

Two structural forces produce the null, and separating them determines what the result does and does not say about the theory.

The first is general: failures are state-neutral. A declined recommendation costs the period and the foregone margin but leaves the state unchanged, so a policy that wants the user at a high-margin category can recommend that category directly; each success both collects the margin and moves the state. Direct targeting is itself a cultivation strategy, and the myopic policy already prices its immediate cost. An intermediate item can beat direct targeting only when the target is effectively unreachable, meaning alignment sensitivity is large enough that a misaligned target's consumption probability is negligible while the intermediate item both consumes well and moves the state toward it.

The second force is geometric. With one-hot category features, consuming category b changes alignment with any other category a by $\rho\,(x_b^\top x_a - z^\top x_a) = -\rho\, z_a < 0$: a click moves the user toward the clicked category and away from every other one, so the cross-item bridge of Jiangze's classification, consume b now so that a different item a becomes attractive later, is impossible by construction; it requires $x_b^\top x_a$ to be substantial, and orthogonal vertices set it to zero. The bridge test therefore needs dense features, and we ran it: items become the eight categories' unit-normalized LSA centroids, whose Gram off-diagonals span 0.44 to 0.89 (mean 0.71; about half of that similarity is a common first LSA component, with the mean falling to roughly 0.35 when it is removed, so the raw figure overstates topical closeness), states are empirical dense history vectors, and (α, κ) come from the dense click model. The result does not change. The aware-vs-myopic gap stays within 0.01% at every ρ, and Meichun's exact two-period bridge test, the share of empirical states whose optimal two-period first action is a persuasive bridge, returns 0.0% in the dense geometry (in the one-hot geometry the share is provably zero: success on i weakly lowers every other category's value, so a bridge can never beat the myopic action over two periods). The diagnostic is not dead: it fires on the same empirical states once κ reaches 6 to 8, exactly where the gate analysis below says steering begins. At MIND's alignment sensitivity, bridges do not exist anywhere in the empirical state distribution.

Quantifying the gate requires care, because raising κ with fixed intercepts raises overall engagement, not just discrimination, and the two must be separated. The raw sweep (Table 4) shows the premium reaching 9.9% at $\kappa = 7$, $\rho = 0.15$; but an iso-engagement sweep on the cluster, re-centering every intercept at each κ so that category-level average click rates match the calibrated baseline, collapses that cell from 6.3% to 1.1% (at $\kappa = 5$: 1.2% to 0.7%). Most of the raw gate was engagement, not discrimination. The honest statement is stronger than the one it replaces: holding engagement fixed, even four times MIND's alignment sensitivity yields a premium of about one percent, so alignment sensitivity alone, at any plausible level, does not make cultivation-aware planning valuable in this model.

Table 4: Aware−blind revenue gap (%) as alignment sensitivity varies, heterogeneous margins, all other parameters calibrated. The first row is the MIND estimate.

κ	gap at $\rho = 0.035$	gap at $\rho = 0.15$
1.75	0.0%	0.1%
3.00	0.0%	0.3%
5.00	0.1%	1.0%
7.00	0.3%	9.9%

4.3 The transition the data prefer

The last experiment simulates the transition the estimation actually supports: a tenure-dependent click step $\rho_t = 1/(n_0 + N_t)$ with n_0 drawn from empirical history lengths, and the backfire update on declined recommendations, with the planner planning through both branches (validated against exact expectimax dynamic programming on a small instance, maximum relative error 0.4%). Backfire breaks state-neutrality, so this is where a dynamic premium could first appear, and the sign of its effect is not obvious in advance: it penalizes failed direct targeting but also makes failed bridge attempts costly.

The answer is a complementarity. At the calibrated $\hat{\kappa} = 1.751$, nothing happens: gaps stay between 0.05% and 0.13% across backfire strengths and user tenures, because damaging a state that barely moves choice probabilities is not much of a threat. At $\kappa = 5$ the same backfire term raises the premium from 0.12% to 4.29%, with the aware policy recommending non-myopically in a quarter of all periods; re-run on the cluster under the corrected state definition, empirical tenures, and ten paired seeds, the effect is 0.31% without backfire and 2.9% with it, so the complementarity survives every specification correction applied so far. Backfire supplies the cost of failure and κ supplies its consequence; either alone is worthless, together they produce the first economically meaningful planning premium in any scenario anchored to the data. The model extension the estimation motivates is therefore not decorative: resistance is the ingredient that makes cultivation strategic, provided alignment sensitivity is high enough to transmit it.

Learning, finally, is a real but symmetric cost. Under heterogeneous margins the learning policies run roughly ten percent below their oracles, yet the cost hits the aware and blind arms equally, so the planning comparison is unaffected. The narrow reading is the right one: eight persistent category intercepts are learned within a few hundred episodes in this simulator, which says little about learning ephemeral article attractiveness, and the ease is partly an artifact of the category aggregation itself.

5 Model variants and the algorithm benchmark

The remaining question is constructive: which modification of the model makes cultivation-aware planning worth having, and does the data support it? We test every candidate raised in the group's discussion, in two environments: Xinyuan's five-item showcase (the bridge regime: $\kappa = 3$, $\rho = 0.45$, a high-margin hard-to-accept target) and the MIND-calibrated environment.

5.1 The showcase under empirical dynamics

Our exact-DP port reproduces the showcase: the optimal policy follows the bridge path (B1, B2, then harvest the target) with a 28.9% premium over reactive myopia, rising to 42.5% under the spillover transition. Two empirical ingredients change the picture. The tenure-declining update $\rho_t = 1/(n_0+t)$, the form the MIND estimation prefers, collapses the premium to 0.0%: the optimal path becomes pure myopia even in the regime built to showcase bridges, because a step size that shrinks with every success removes the compounding that cultivation requires. And backfire, which creates the premium at low click probabilities, taxes it here (25.5% at ten times the estimated strength): when bridges are the plan, failed bridge attempts damage the state the plan depends on. MIND-calibrated parameters in the same geometry give 0.0%. The showcase is internally consistent, and each data-motivated ingredient moves it toward the null.

5.2 Variants at calibrated parameters

Three model variants proposed as repairs, each simulated aware-versus- myopic under common random numbers at the calibrated (κ, ρ): a terminal-payoff objective in which revenue accrues only in the final five periods, so early periods are free cultivation (gap 0.22%); distance-based acceptance $p = \sigma(\alpha_c - \lambda\|z - e_c\|^2)$ with intercepts recentered to hold engagement fixed, which hardens the reachability wall (gap 0.16%); and a slate model in which the platform recommends three categories and the user chooses by MNL with an outside option calibrated to the observed click rate (gap 0.01%). None exceeds a quarter of one percent. The spillover transition, the one variant that is directly estimable, is rejected by the data: profiling $z^+ = z + \rho[(1-s)I + sS](e_c - z)$ with S built from the content similarity of categories gives $\hat{s} = 0.0$, with the likelihood falling monotonically in s (by 49 points at $s = 0.3$). The conclusion of Section 4 is not an artifact of the objective, the acceptance form, the choice model, or a missing spillover channel.

5.3 The index policy

The planned paper centers on the exploitation-cultivation-exploration index. Our implementation reproduces the note's worked example exactly (the myopic and dynamic scores and the bridge-to-target path). The learning benchmark in the showcase regime, with intercepts unknown and learned across 400 episodes, is sobering: estimate-then-plug-in dynamic programming attains 99.3% of the full-information oracle, while the approximate index with the myopic-payoff potential ranges from 75.6% to 78.8%-level performance depending on tuning, and the best configuration found (geometric potential toward the target, horizon- scaled cultivation weight, small exploration bonus) reaches 95.5%. Two implications for the paper. First, the benchmark the heuristic must beat is not myopia but plug-in planning, which is already near-oracle when a few persistent intercepts are the only unknowns; the index's case must rest on settings where plug-in fails, such as state uncertainty or short-lived items. Second, the heuristic's performance is dominated by the choice of potential Φ and the cultivation weight η_h, which the theory currently leaves free; pinning them down is the highest-value theoretical task.

6 Summary

1. **The demand form and short-run state dependence receive strong predictive support.** Alignment drives clicks ($\hat{\kappa} = 1.751$) and recent consumption updates an effective preference state ($\hat{\rho} = 0.035$, half-life about 19 clicks). Both are behavioral quantities under the incumbent recommender; neither is yet a causal taste primitive, and the constant-step linear transition is an adequate local description rather than a validated law.

2. **The data motivate two specific extensions.** Exposure without a click is negatively associated with subsequent affinity, and the update rate declines with tenure along the Bayesian-updating signature $1/(N + T/2)$. Resistance and tenure-dependence are not robustness footnotes; they are the empirically supported route to a model in which cultivation has bite, and they concentrate whatever cultivation exists among new users.

3. **At calibrated parameters, planning adds nothing beyond reactive myopia, and the result is now thorough.** The null survives high engagement, long horizons, a systematic search over 10,000 margin configurations (99th-percentile gap 0.50%), and the move from one-hot to dense content geometry, where the exact two-period test finds bridges for zero percent of empirical states. Two mechanisms explain it: failures are state-neutral, so direct targeting is itself the best cultivator, and alignment sensitivity alone cannot rescue planning: at fixed engagement, even four times MIND's sensitivity yields about a one percent premium. The paired design turns the null into an equivalence result, with a 95% upper bound of 0.53% on the premium across all main scenarios.

4. **The premium exists, and the data point at its recipe.** Simulating the transition the estimation prefers shows backfire and alignment sensitivity are complements: at $\hat{\kappa}$ backfire changes nothing, at $\kappa = 5$ it moves the premium from about 0.31% to 2.9% (cluster rerun, corrected states, empirical tenures) with a quarter of recommendations non-myopic. Resistance makes cultivation strategic; alignment sensitivity transmits it. That is a concrete, falsifiable claim about which platforms should invest in cultivation-aware algorithms.

5. **No proposed repair changes the calibrated answer, and the data reject the one testable repair.** Terminal payoffs, distance-based acceptance, and MNL slates all leave the premium below a quarter percent; the spillover transition is estimated at $\hat{s} = 0.0$. In the showcase regime the empirically preferred tenure-declining update eliminates the bridge path outright. And the algorithm benchmark shows plug-in planning at 99.3% of oracle, so the index heuristic's contribution must be established against plug-in, not against myopia, and hinges on disciplining its free potential and cultivation weight.

6. **Standing caveats.** Category-level states compress articles to 18 types, estimated for 14 and simulated for the largest 8 with the state renormalized; MIND has no margins; the 14-day window bounds expressible cultivation; position effects and within-impression competition are not modeled; likelihood statistics treat impression items as independent while they cluster within users (a cluster bootstrap keeps every conclusion at roughly a third of the nominal statistics); and both estimated primitives are policy-dependent, the right inputs for an incumbent-platform calibration and the wrong ones to extrapolate across policies.